

Terminal Sliding Mode Control (TSMC) based Cooperative Load Transportation using Multiple Drones

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Abstract—Objective of this paper is to design a suitable control algorithm for cooperative transportation of an object using multiple drones based on decentralized control technique. A terminal sliding mode control (TSMC) based algorithm for motion tracking of a system of multiple Quad-rotor type Unmanned Aerial Vehicle (UAV) is proposed. TSMC is a novel class of sliding mode control giving rise to a finite time convergence of reduced order dynamics, delivering fast response, high precision, and strong robustness. A TSMC based robust cooperative control strategy is proposed for the system with multiple drones and individual control vectors are defined for controlling the combined system in presence of wind disturbances. The finite time convergence of control laws are proved using homogeneous Lyapunov function. The simulation results demonstrate the effectiveness and superior performance over conventional sliding mode-based technique for the cooperative quad rotors to transport a common payload.

Keywords—cooperative control, drones, UAV, sliding mode control, terminal sliding mode control, load transportation using drones, finite time convergence.

I. INTRODUCTION

The word ‘drone’ is very popular now a days. They are being used in various technological sectors. Recently they are being considered as future load transporting vehicles. Various MNC’s are already testing delivery using drones. But for large scale load transportation multiple drones are required and there lies the importance of cooperative control of drones. Cooperative control between robots, robot and human are already very popular. Cooperative and decentralized controlling techniques for drones are discussed in the works [1],[7],[8],[9].

There are various methods or control techniques for controlling a drone. The dynamics of drone are highly affected due to aerial disturbances and cross coupling effects due to translational and rotational motion. For this reason a robust control system is required which can handle this non linearities and dynamic stability problem. Sliding Mode Control(SMC) is quite popular among them because it is quite robust towards uncertainties and disturbances. The importance and application of sliding mode control are well described in the works [10],[12]. But sliding mode control provides asymptotic convergence and thus has very slow response. So to overcome this sliding mode control has been modified with a terminal attractor and named as Terminal SMC(TSMC). TSMC provides finite time convergence, and thus also provides fast convergence compared to normal SMC. The nature of TSMC are documented in the works [5],[6],[11],[13].

In this work, we developed a fast and robust cooperative aerial load transport system. For this decentralized cooperative control laws for a group of drones in a formation are developed where each vehicle is controlled individually. The modelling and behavior of cooperative systems are documented in the works [1], [2], [7], [8], [9]. Moore Penrose optimization theory is applied for robust modelling of cooperative system.

In this paper we have developed a TSMC based decentralized cooperative system for a system of four quad rotor type drones and a common payload. The proposed controller can successfully handle external disturbances for the cooperative system and also provides very fast convergence in finite time. The organization of this paper I as follows. In Section II,

mathematical modelling of drone is established. In Section III, the cooperative control strategy is discussed. Section IV focusses on derivation of control vectors for individual drones. In Section V, simulation results are presented in the presence of disturbance and system uncertainties. Section VI discusses the conclusions and future scopes of this project.

II. MATHEMATICAL MODELLING OF DRONE

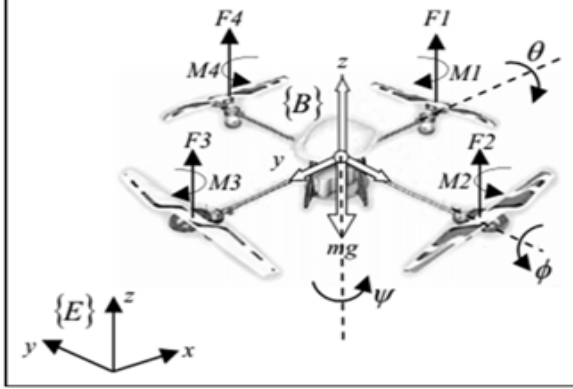


Figure 1. Configuration, Inertial and body reference frame

Quadrotor has 6 degree of freedom $[x, y, z, \theta, \Phi, \Psi]$, 3 translation and 3 rotational respectively. x, y, z denote the body frame axes, Φ denotes roll angle, θ denotes pitch angle and Ψ denotes yaw angle in body frame (Drone body frame). Forces acting on body frame are to be mapped on inertial earth reference frame using Euler Rotation Matrices[3].

$$\begin{aligned}
 R_\Phi &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\Phi) & \sin(\Phi) \\ 0 & \sin(\Phi) & \cos(\Phi) \end{bmatrix} \\
 R_\theta &= \begin{bmatrix} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \\
 R_\Psi &= \begin{bmatrix} \cos(\Psi) & \sin(\Psi) & 0 \\ -\sin(\Psi) & \cos(\Psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 R_B^E &= \begin{bmatrix} c\theta c\Psi & s\theta s\theta c\Psi - c\theta s\Psi & -c\theta s\theta c\Psi + s\theta s\Psi \\ -c\theta s\Psi & -s\theta s\theta s\Psi + c\theta c\Psi & c\theta s\theta s\Psi + s\theta c\Psi \\ s\theta & -s\theta c\theta & c\theta c\theta \end{bmatrix} \\
 \begin{bmatrix} i_E \\ j_E \\ k_E \end{bmatrix} &= R_B^E \begin{bmatrix} i_b \\ j_b \\ k_b \end{bmatrix} \quad (1)
 \end{aligned}$$

Total Lift force generated by rotors given by $U_1 = F_1 + F_2 + F_3 + F_4$, where F_i are the thrust force generated by i th rotor. $F_T^b = [0 \ 0 \ U_1]^T$, lift force given with respect to body frame.

Forces with respect to earth frame, $F = R_B^E * F_T^b$ (3)

Translational dynamics of quadrotor are given by following equations:

$$\begin{aligned}
 \ddot{x} &= \frac{U_1(\sin(\theta)\cos(\phi)\cos(\Psi) + \sin(\phi)\sin(\Psi))}{m} + w_x \\
 \ddot{y} &= \frac{U_1(\sin(\theta)\cos(\phi)\sin(\Psi) - \sin(\phi)\cos(\Psi))}{m} + w_y \\
 \ddot{z} &= \frac{U_1(\cos(\theta)\cos(\phi)) - mg}{m} + w_z \quad (2)
 \end{aligned}$$

$\ddot{x}, \ddot{y}, \ddot{z}$ are the acceleration of drone with respect to Earth frame due to thrust control U_1 . w_x, w_y, w_z are air gust disturbances, m is the mass of drone, and g is the gravitational constant (9.8 m/s²). Rotational dynamics of drone are given by following equations:

$$I_x \dot{p} = (I_y - I_z)qr - J_r q\Omega + LU_2$$

$$I_y \dot{q} = (I_z - I_x)pr - J_r p\Omega + LU_3$$

$$I_z \dot{r} = (I_x - I_y)qp + LU_4$$

L is the arm length of quad rotor. I_x, I_y, I_z are the moments of inertia about x, y, z axis respectively. p, q, r are the angular speeds on body frame. LU_2, LU_3, LU_4 are the moments about x, y and z axis respectively. J_r is the moment of inertia of each propeller. The inputs U_2, U_3, U_4 are related to the rotations of the quad rotor. So final rotational dynamics are given by following equations:

$$\ddot{\Phi} = \frac{(I_y - I_z)\dot{\theta}\dot{\Psi} + LU_2 - J_r \dot{\theta}\Omega}{I_x} + w_\Phi$$

$$\ddot{\theta} = \frac{(I_z - I_x)\dot{\Phi}\dot{\Psi} + LU_3 - J_r \dot{\Phi}\Omega}{I_y} + w_\theta$$

$$\ddot{\Psi} = \frac{(I_x - I_y)\dot{\theta}\dot{\Phi} + LU_4}{I_z} + w_\Psi$$

w_Φ, w_θ, w_Ψ are the air gust disturbances. Ω is the overall residual angular speed. $\Omega = \Omega_2 + \Omega_3 - \Omega_1 - \Omega_4$, where Ω_i are the angular speed of i th rotor. Relation of Ω with control vectors are given by following,

$$\begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix} = \begin{bmatrix} \frac{1}{4b} & -\frac{1}{2bl} & 0 & \frac{1}{4b} \\ \frac{1}{4b} & 0 & -\frac{1}{2bl} & -\frac{1}{4d} \\ \frac{1}{4b} & \frac{1}{2bl} & 0 & \frac{1}{4d} \\ \frac{1}{4b} & 0 & -\frac{1}{2bl} & -\frac{1}{4d} \end{bmatrix} \quad (4)$$

Here, d is thrust coefficient and b is drag coefficient. Therefore, the desired motor speed can be calculated and corresponding input can be sent to motor controllers. The drone model can be written in state space form $\dot{X} = f(X, U)$ by introducing $X = [x_1, x_2, \dots, x_{11}, x_{12}]$, where

$$[x_1 = \Phi, x_2 = \dot{\Phi}, x_3 = \theta, x_4 = \dot{\theta}, x_5 = \Psi, x_6 = \dot{\Psi}, x_7 = x, x_8 = \dot{x}, x_9 = y, x_{10} = \dot{y}, x_{11} = z, x_{12} = \dot{z}]$$

Assumption 1: It is assumed that roll, pitch and yaw angle satisfy the conditions, $\Phi \leq \pi/2, \theta \leq \pi/2, \Psi \leq \pi$ for $t > 0$.

III. MATHEMATICAL MODELLING OF COUPLED SYSTEM

Figure 2 shows a team of four quad rotor drones manipulate a cross configuration payload in three dimensions. The coordinate systems include the world frame W , and body frame B . The body frame axes are considered as the principle axes of the entire system and each quad rotor has an individual body frame Q_i , where $i = \{1, 2, 3, \dots, n\}$ and n is the number of drones in the system. Here $x-y-z$ Euler angles are used to define roll, pitch and yaw angles.

The rotation matrix B to W is given by R_B . Center of mass of i th drone is considered (x_{Qi}, y_{Qi}, z_{Qi}) .

Assumptions for quad rotor drones:

- 1)The payload is rigid and completely homogeneous, has a cross configuration structure and the center of mass of the object is intended as the center of mass of the coupled system.
- 2)The mass of payload is sufficiently small such that four drones are able to lift the object.
- 3)The payload does not flip during manipulation.

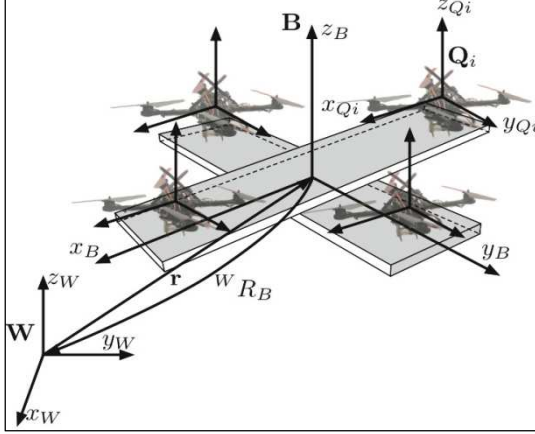


Figure 2. Scheme of payload transportation using 4 drones

The motion equation of each quad rotor is given by,

$$m_i \ddot{z}_{Qi} = -m_i g + F_{Bi} \quad (5)$$

Here m_i is the mass of i th quad rotor, g is gravity and F_{Bi} is the lift force generated by each quad rotor. Also by Newton's 2nd law

$$m_L \ddot{z} = -m_L g \quad (6)$$

Where m_L is the mass of the payload. Now as all the forces are used for lifting and moving toward positive Z direction, the equation of motion for the coupled system can be given as,

$$m \ddot{z} = -mg + F_B \quad (7)$$

F_B is the total lift force of all quad rotors and m is the sum of drones and object's mass, $m = m_1 + m_2 + m_3 + m_4$. The acceleration of center of mass is given by, $m \ddot{r} = -mg + R_B F_B$. r is the position vector to the center of mass of i th drone.

By considering forces and moments generated by each UAV, a relationship between the behaviour of system and the agents has to be developed. Finally the equation of motion of coupled system is obtained as,

$$\begin{bmatrix} F_B \\ M_{XB} \\ M_{YB} \\ M_{ZB} \end{bmatrix} = \sum_i \begin{bmatrix} 1 & 0 & 0 & 0 \\ y_i & \cos(\psi_i) & -\sin(\psi_i) & 0 \\ -x_i & \sin(\psi_i) & \cos(\psi_i) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} F_{Qi} \\ M_{XQi} \\ M_{YQi} \\ M_{ZQi} \end{bmatrix}$$

Where M_{XB} , M_{YB} , M_{ZB} are the body moments in Z_{Qi} direction and F_{Qi} is the total force of each quadrotor, and M_{XQi} , M_{YQi} , M_{ZQi} are the moments around each quad rotor body frame axis.

IV. COOPERATIVE CONTROL STRATEGY

This control algorithm is a decentralized control algorithm and includes three parts. In the first part of this the control fundamental vectors for n quad rotors that define a pseudo inverse matrix of the mathematical theory of optimal control using Moore - Penrose are determined. The second part consists of determining individual control vectors by TSMC. Finally in the last part the cooperative control algorithm can be established by combining the individual control vectors and fundamental control vectors.

A. Control Vectors of Cooperative Control Law

The derivation of these control vectors are well described in the works [1],[2],[7]. From there the cooperative strategy is established. By using all force vectors of the coupled system in linear system (), four equations for four quad rotor drones can be described as,

$$[F_B, M_{XB}, M_{YB}, M_{ZB}]^T = Xu \quad (8)$$

Where $X \in R^{4 \times 4n}$, is a constant vector containing the states of translational and rotational subsystems, and $u \in R^{4n}$ includes four control input vectors for n quad rotors,

$$u = [F_{Bq1}, M_{XBq1}, M_{YBq1}, M_{ZBq1}, \dots, F_{Bqn}, M_{XBqn}, M_{YBqn}, M_{ZBqn}]^T \quad (9)$$

The cost function to minimize control inputs is given by,

$$J = \sum_i (\Lambda_{Fi} F_{qi}^2 + \Lambda_{Mxi} F_{xqi}^2 + \Lambda_{Myi} F_{yqi}^2 + \Lambda_{Mzi} M_{zqi}^2) \quad (10)$$

in which, Λ_{Fi} , Λ_{Mxi} , Λ_{Myi} , Λ_{Mzi} are the weight of each of the control inputs. By considering the features of theory of Moore - Penrose inverse, pointwise minimization of J will be done and optimal control vector can be written as,

$$u^* = \Gamma^{-2} X^T (X \Gamma^{-2} X^T)^{-1} [F_B^{des}, M_{XB}^{des}, M_{YB}^{des}, M_{ZB}^{des}]^T \quad (11)$$

$$\Gamma = [\sqrt{\Lambda_{F1}}, \sqrt{\Lambda_{Mx1}}, \sqrt{\Lambda_{My1}}, \sqrt{\Lambda_{Mz1}}, \dots, \sqrt{\Lambda_{Fn}}, \sqrt{\Lambda_{Mxn}}, \sqrt{\Lambda_{Myn}}, \sqrt{\Lambda_{Mzn}}]$$

By assuming the weight of basic control vectors,

$$\Lambda_{Fi} = \Lambda_F, \Lambda_{Mxi} = \Lambda_{Myi} = \Lambda_{Mxy} \quad (12)$$

$$\Gamma^{-2} X^T (X \Gamma^{-2} X^T)^{-1} = [U_{Fqn}, U_{Mxqn}, U_{Myqn}]$$

Assumption 2: While forming the control law we have assumed the X and Y coordinates of the centroid of the object and centroid of quad rotors coincide according to Figure 2. The following assumptions were made, $\sum x_i = \sum y_i = 0$, $\sum x_i y_i = 0$.

Also, since all the quad rotor drones have a common role in lifting the object off the ground and equal partnership, the body force F and yaw produced by each of them is almost equal. Thus the total force and yaw moment, roll moment and pitch moment of the coupled system are obtained as follows,

$$U_{Fqn} = \frac{1}{n} [1, 0, 0, 0, \dots, 1, 0, 0, 0]^T$$

$$U_{Mzqn} = \frac{1}{n} [0, 0, 0, 1, \dots, 0, 0, 0, 1]^T \quad (13)$$

$$U_{Mxqn} = \frac{1}{\frac{\Lambda_{Mxy}}{\Lambda_F} \sum y_i^2 + n} \left[\frac{\Lambda_{Mxy}}{\Lambda_F} y_{q1}, C\psi_{q1}, S\psi_{q1}, 0, \dots, \frac{\Lambda_{Mxy}}{\Lambda_F} y_{qn}, C\psi_{qn}, S\psi_{qn}, 0 \right]^T$$

$$U_{Myqn} = \frac{1}{\frac{\Lambda_{Mxy}}{\Lambda_F} \sum x_i^2 + n} \left[-\frac{\Lambda_{Mxy}}{\Lambda_F} x_{q1}, -S\Psi_{q1}, C\Psi_{q1}, 0, \dots, -\frac{\Lambda_{Mxy}}{\Lambda_F} x_{qn}, -S\Psi_{qn}, C\Psi_{qn}, 0 \right]^T$$

It is important to notice to the increment of the force produced by each n quad rotor drones will lead to decrement of the individual body moments generated by them, according to the factor $\frac{\Lambda_{Mxy}}{\Lambda_F}$. In next step individual control vectors $[F_B^{des}, M_{xB}^{des}, M_{yB}^{des}, M_{zB}^{des}]^T$ are to be found and the cooperative control vectors for n collaborative drones is obtained as follows,

$$U_{coop|qn} = U_{Fqn} F_{Bn}^{des} + U_{Mxqn} M_{xBn}^{des} + U_{Myqn} M_{yBn}^{des} + U_{Mzqn} M_{zBn}^{des} \quad (14)$$

B. Derivation of Individual Control Vectors Based on TSMC

In this section we will derive individual control vectors based on terminal sliding mode control. The control law consists of two parts, one continuous part which is to be derived from sliding equation and second part is discontinuous switching function which reduces chattering phenomena.

$$U_{\tilde{X}} = U_{\tilde{X}con} + U_{\tilde{X}dis} \quad (15)$$

Where $\tilde{X}=[x, y, z, \Phi, \theta, \Psi]$. The discontinuous part is given by $U_{dis} = k_D \frac{s}{|s|+\delta}$, where k_D and δ are tuning parameters [14].

Now we first derive the continuous part of control law. The sliding surface equations are given by,

$$\begin{aligned} S_x &= \dot{e}_x + \lambda e_x + b * \text{sign}(e_x)^{\frac{p}{q}} \\ S_y &= \dot{e}_y + \lambda e_y + b * \text{sign}(e_y)^{\frac{p}{q}} \\ S_z &= \dot{e}_z + \lambda e_z + b * \text{sign}(e_z)^{\frac{p}{q}} \\ S_\Phi &= \dot{e}_\Phi + \lambda e_\Phi + b * \text{sign}(e_\Phi)^{\frac{p}{q}} \\ S_\theta &= \dot{e}_\theta + \lambda e_\theta + b * \text{sign}(e_\theta)^{\frac{p}{q}} \\ S_\Psi &= \dot{e}_\Psi + \lambda e_\Psi + b * \text{sign}(e_\Psi)^{\frac{p}{q}} \end{aligned} \quad (16)$$

$$\begin{aligned} e_{\tilde{X}} &= \tilde{X}_D - \tilde{X} \\ \text{sign}(e_{\tilde{X}})^{\frac{p}{q}} &= \text{abs}(e_{\tilde{X}})^{\frac{p}{q}} * \text{sign}(e_{\tilde{X}}) \end{aligned} \quad (17)$$

Here λ, b are positive constants and p and q are positive integers, where $p < q$. The time derivative of sliding surface is given by,

$$\dot{S}_{\tilde{X}} = \ddot{e}_{\tilde{X}} + \lambda \dot{e}_{\tilde{X}} + b * \frac{p}{q} * \text{sign}(e_{\tilde{X}}) * \text{abs}(e_{\tilde{X}})^{\frac{(p-1)}{q}} * \dot{e}_{\tilde{X}} \quad (18)$$

By generating sliding mode reaching law [4],

$$\dot{S}_{\tilde{X}} = -\xi_{\tilde{X}} S_{\tilde{X}} - \eta_{\tilde{X}} \text{sign}(S_{\tilde{X}}) \quad (19)$$

And replacing the acceleration equations, the control inputs are obtained as follows,

$$U_{1con} = \left[g + \left(\lambda + b * \frac{p}{q} * \text{abs}(e_z)^{\frac{(p-1)}{q}} * \text{sign}(e_z) \right) \dot{e}_z + \ddot{z}_D + \xi_z S_z + \eta_z \text{sign}(S_z) - w_z \right] \frac{m}{\mu_3}$$

$$\begin{aligned} U_{2con} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_\Phi)^{\frac{(p-1)}{q}} * \text{sign}(e_\Phi) \right) (\dot{\Phi}_D - \dot{\Phi}) + \ddot{\Phi}_D - \dot{\theta} \dot{\Psi} \left(\frac{I_y - I_z}{I_x} \right) + \xi_\Phi S_\Phi + \eta_\Phi \text{sign}(S_\Phi) - w_\Phi \right] \frac{I_x}{I} \\ U_{3con} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_\theta)^{\frac{(p-1)}{q}} * \text{sign}(e_\theta) \right) (\dot{\theta}_D - \dot{\theta}) + \ddot{\theta}_D - \dot{\Phi} \dot{\Psi} \left(\frac{I_z - I_x}{I_y} \right) + \xi_\theta S_\theta + \eta_\theta \text{sign}(S_\theta) - w_\theta \right] \frac{I_y}{I} \\ U_{4con} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_\Psi)^{\frac{(p-1)}{q}} * \text{sign}(e_\Psi) \right) (\dot{\Psi}_D - \dot{\Psi}) + \ddot{\Psi}_D - \dot{\theta} \dot{\Phi} \left(\frac{I_x - I_y}{I_z} \right) + \xi_\Psi S_\Psi + \eta_\Psi \text{sign}(S_\Psi) - w_\Psi \right] \frac{I_z}{I} \\ U_{Xcon} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_X)^{\frac{(p-1)}{q}} * \text{sign}(e_X) \right) \dot{e}_X + \ddot{X}_D + \xi_X S_X + \eta_X \text{sign}(S_X) - w_X \right] \frac{m}{\mu_1} \\ U_{Ycon} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_Y)^{\frac{(p-1)}{q}} * \text{sign}(e_Y) \right) \dot{e}_Y + \ddot{Y}_D + \xi_Y S_Y + \eta_Y \text{sign}(S_Y) - w_Y \right] \frac{m}{\mu_2} \end{aligned} \quad (20)$$

Where μ_1, μ_2, μ_3 are function of Euler angles and are given by,

$$\begin{aligned} \mu_1 &= \cos(\Phi) \sin(\theta) \cos(\Psi) + \sin(\Phi) \sin(\Psi) \\ \mu_2 &= \cos(\Phi) \sin(\theta) \sin(\Psi) - \sin(\Phi) \sin(\Psi) \\ \mu_3 &= \cos(\Phi) \cos(\theta) \end{aligned} \quad (21)$$

Remarks 1: In the above equations term containing moment of inertia of rotor (J_r) has been removed. Compared to the moment of drone body, the moment of propeller is negligible.

Now for proving stability of the control law, a Lyapunov function is defined as, $V = \frac{1}{2} S^2$. For stable system the time derivative of the function must be negative definite. Now, for altitude control, $\dot{V} = S_Z \dot{S}_Z$ must be $\dot{V} < 0$.

$$\begin{aligned} \dot{V} &= S_Z (-\xi_Z S_Z - \eta_Z \text{sign}(S_Z)) \\ \dot{V} &= -\xi_Z S_Z^2 - \eta_Z |S_Z| \end{aligned} \quad (22)$$

ξ_Z and η_Z are positive tuning constants, so for all non zero S_Z , $\dot{V} < 0$. Now, equivalent control law consists of continuous and discontinuous part and is given by,

$$\begin{aligned} U_{1eq} &= \left[g + \left(\lambda + b * \frac{p}{q} * \text{abs}(e_z)^{\frac{(p-1)}{q}} * \text{sign}(e_z) \right) \dot{e}_z + \ddot{z}_D + \xi_z S_z + \eta_z \text{sign}(S_z) - w_z \right] \frac{m}{\mu_3} + k_{DZ} \frac{S_z}{|S_z| + \delta} \\ U_{1eq} &= U_{1con} + U_{1dis} \end{aligned} \quad (23)$$

Now for designing discontinuous signal, we take a Lyapunov function $V1 = \frac{1}{2} S_Z^2$. For stable control law,

$$\begin{aligned} \dot{V1} &= S_Z \dot{S}_Z \\ \dot{V1} &= S_Z (\dot{e}_z + \lambda e_z + b * \frac{p}{q} * \text{sign}(e_z) * \text{abs}(e_z)^{\frac{(p-1)}{q}} * \dot{e}_z) \\ \dot{V1} &= S_Z (\ddot{z}_D - \frac{U_{1eq} * \mu_3 - mg}{m} - w_z + \lambda \dot{e}_z + b * \frac{p}{q} * \text{sign}(e_z) * \text{abs}(e_z)^{\frac{(p-1)}{q}} * \dot{e}_z) \end{aligned}$$

$$\dot{V}_1 = S_Z \left(-k_{DZ} * \frac{S_Z}{|S_Z| + \delta} * \frac{\mu_3}{m} \right) \quad (24)$$

So, for $k_D > 0$, for all non zero S_Z , $\dot{V} < 0$. The same stability can be proved for altitude control also [14]. So finally the individual control vectors are given as follows,

$$\begin{aligned} F_{Bn}^{des} &= \left[g + \left(\lambda + b * \frac{p}{q} * \text{abs}(e_{zn})^{\frac{p-1}{q}} * \text{sign}(e_{zn}) \right) \ddot{e}_z + \ddot{z}_D + \xi_Z S_{zn} \right. \\ &\quad \left. + \eta_Z \text{sign}(S_{zn}) - w_z \right] \frac{m}{\mu_{3n}} + k_{DZ} \frac{S_{zn}}{|S_{zn}| + \delta} \\ M_{XBn}^{des} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_{\phi n})^{\frac{p-1}{q}} * \text{sign}(e_{\phi n}) \right) (\ddot{\phi}_D - \ddot{\phi}_n) + \ddot{\phi}_D \right. \\ &\quad \left. - \ddot{\theta}_n \psi_n \left(\frac{I_y - I_z}{I_x} \right) + \xi_\phi S_{\phi n} + \eta_\phi \text{sign}(S_{\phi n}) - w_\phi \right] \frac{I_x}{I} \\ &\quad + k_{D\phi} \frac{S_{\phi n}}{|S_{\phi n}| + \delta} \\ M_{YBn}^{des} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_{\theta n})^{\frac{p-1}{q}} * \text{sign}(e_{\theta n}) \right) (\ddot{\theta}_D - \ddot{\theta}_n) + \ddot{\theta}_D \right. \\ &\quad \left. - \ddot{\phi}_n \psi_n \left(\frac{I_z - I_x}{I_y} \right) + \xi_\theta S_{\theta n} + \eta_\theta \text{sign}(S_{\theta n}) - w_\theta \right] \frac{I_y}{I} \\ &\quad + k_{D\theta} \frac{S_{\theta n}}{|S_{\theta n}| + \delta} \\ M_{ZBn}^{des} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_{\psi n})^{\frac{p-1}{q}} * \text{sign}(e_{\psi n}) \right) (\ddot{\psi}_D - \ddot{\psi}_n) + \ddot{\psi}_D \right. \\ &\quad \left. - \ddot{\theta}_n \phi_n \left(\frac{I_x - I_y}{I_z} \right) + \xi_\psi S_{\psi n} + \eta_\psi \text{sign}(S_{\psi n}) - w_\psi \right] \frac{I_z}{I} \\ &\quad + k_{D\psi} \frac{S_{\psi n}}{|S_{\psi n}| + \delta} \\ U_{Xeq} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_{xn})^{\frac{p-1}{q}} * \text{sign}(e_{xn}) \right) \ddot{e}_{xn} + \ddot{x}_D + \xi_X S_{xn} \right. \\ &\quad \left. + \eta_X \text{sign}(S_{xn}) - w_x \right] \frac{m}{\mu_{1n}} + k_{DX} \frac{S_{xn}}{|S_{xn}| + \delta} \\ U_{Yeq} &= \left[\left(\lambda + b * \frac{p}{q} * \text{abs}(e_{yn})^{\frac{p-1}{q}} * \text{sign}(e_{yn}) \right) \ddot{e}_{yn} + \ddot{y}_D + \xi_Y S_{yn} \right. \\ &\quad \left. + \eta_Y \text{sign}(S_{yn}) - w_y \right] \frac{m}{\mu_{2n}} + k_{DY} \frac{S_{yn}}{|S_{yn}| + \delta} \quad (25) \end{aligned}$$

Where,

$$\begin{aligned} \mu_1 &= \cos(\Phi_n) \sin(\theta_n) \cos(\Psi_n) + \sin(\Phi_n) \sin(\Psi_n) \\ \mu_2 &= \cos(\Phi_n) \sin(\theta_n) \sin(\Psi_n) - \sin(\Phi_n) \sin(\Psi_n) \\ \mu_3 &= \cos(\Phi_n) \cos(\theta_n) \end{aligned} \quad (26)$$

C. Final Cooperative Control Vectors

Now combining the control fundamental vectors in eqns (13) with individual control vectors in eqns (25), the derived cooperative control vectors are given by following,

$$\begin{aligned} &\begin{pmatrix} U_{coop|q1} \\ U_{coop|q2} \\ U_{coop|q3} \\ U_{coop|q4} \end{pmatrix} \\ &= \begin{pmatrix} U_{Fq1} F_{Bq1}^{des} + U_{MXq1} M_{XBq1}^{des} + U_{MYq1} M_{YBq1}^{des} + U_{MZq1} M_{ZBq1}^{des} \\ U_{Fq1} F_{Bq2}^{des} + U_{MXq1} M_{XBq2}^{des} + U_{MYq1} M_{YBq2}^{des} + U_{MZq1} M_{ZBq2}^{des} \\ U_{Fq1} F_{Bq3}^{des} + U_{MXq1} M_{XBq3}^{des} + U_{MYq1} M_{YBq3}^{des} + U_{MZq1} M_{ZBq3}^{des} \\ U_{Fq1} F_{Bq4}^{des} + U_{MXq1} M_{XBq4}^{des} + U_{MYq1} M_{YBq4}^{des} + U_{MZq1} M_{ZBq4}^{des} \end{pmatrix} \quad (26) \end{aligned}$$

To test the robustness of controller against external disturbances, we simulate wind effects as sinusoidal velocity function against the motion of drones. The wind velocity $w(t)$ is given as follows,

$$\begin{bmatrix} w_X(t) \\ w_Y(t) \\ w_Z(t) \end{bmatrix} = \begin{bmatrix} 0.5 \sin\left(\frac{\pi(t-20)}{20}\right) + 0.2 \sin\left(\frac{\pi(t-20)}{10}\right) + 0.06 \sin\left(\frac{\pi(t-20)}{5}\right) \\ 0.5 \sin\left(\frac{\pi(t-30)}{20}\right) + 0.2 \sin\left(\frac{\pi(t-30)}{10}\right) + 0.06 \sin\left(\frac{\pi(t-30)}{5}\right) \\ 0.5 \sin\left(\frac{\pi(t-40)}{20}\right) + 0.2 \sin\left(\frac{\pi(t-40)}{10}\right) + 0.06 \sin\left(\frac{\pi(t-40)}{5}\right) \end{bmatrix} \quad (27)$$

D. Finite Time Convergence of TSMC

TSMC sliding surface equation is defined as ,

$$s = \dot{e} + \lambda e + b * \text{sig}(e)^{\frac{p}{q}} \quad (28)$$

Where λ, b are positive constants and p and q are positive integers such that $p < q$.

$$\dot{e} = -\lambda e - b * \text{abs}(e)^{\frac{p}{q}} * \text{sign}(e) \quad (29)$$

Considering t_2 the time from initial non zero state error $e(t_0)$ to $e(t_2)=0$,

$$\begin{aligned} \int_{e(t_0)}^{e(t_2)} \frac{de}{-\lambda e - b * \text{abs}(e)^{\frac{p}{q}} * \text{sign}(e)} &= \int_{t_0}^{t_2} dt \\ t_2 &= \frac{1}{\lambda \left(1 - \frac{p}{q}\right)} * \ln \left(\frac{\lambda * \text{abs}(e(t_0))^{(1-\frac{p}{q})} + b}{b} \right) \quad (30) \end{aligned}$$

Here t_2 does not tend to infinite, so convergence is proved.

E. Derivation of Individual Control Vectors Based on SMC

For comparing the results of TSMC with conventional SMC, we also define SMC control vectors. The sliding surface of SMC is given by,

$$S_X = \dot{e}_X + \lambda e_X \quad (31)$$

The control vectors will be given by,

$$\begin{aligned} F_{Bn}^{des} &= [g + \lambda \ddot{e}_z + \ddot{z}_D + \xi_Z S_{zn} + \eta_Z \text{sign}(S_{zn}) - w_z] \frac{m}{\mu_{3n}} + k_{DZ} \frac{S_{zn}}{|S_{zn}| + \delta} \\ M_{XBn}^{des} &= \left[\lambda (\ddot{\phi}_D - \ddot{\phi}_n) + \ddot{\phi}_D - \ddot{\theta}_n \psi_n \left(\frac{I_y - I_z}{I_x} \right) + \xi_\phi S_{\phi n} + \eta_\phi \text{sign}(S_{\phi n}) \right. \\ &\quad \left. - w_\phi \right] \frac{I_x}{I} + k_{D\phi} \frac{S_{\phi n}}{|S_{\phi n}| + \delta} \\ M_{YBn}^{des} &= \left[\lambda (\ddot{\theta}_D - \ddot{\theta}_n) + \ddot{\theta}_D - \ddot{\phi}_n \psi_n \left(\frac{I_z - I_x}{I_y} \right) + \xi_\theta S_{\theta n} + \eta_\theta \text{sign}(S_{\theta n}) \right. \\ &\quad \left. - w_\theta \right] \frac{I_y}{I} + k_{D\theta} \frac{S_{\theta n}}{|S_{\theta n}| + \delta} \\ M_{ZBn}^{des} &= \left[\lambda (\ddot{\psi}_D - \ddot{\psi}_n) + \ddot{\psi}_D - \ddot{\theta}_n \phi_n \left(\frac{I_x - I_y}{I_z} \right) + \xi_\psi S_{\psi n} + \eta_\psi \text{sign}(S_{\psi n}) \right. \\ &\quad \left. - w_\psi \right] \frac{I_z}{I} + k_{D\psi} \frac{S_{\psi n}}{|S_{\psi n}| + \delta} \\ U_{Xeq} &= [\lambda \ddot{e}_{xn} + \ddot{x}_D + \xi_X S_{xn} + \eta_X \text{sign}(S_{xn}) - w_x] \frac{m}{\mu_{1n}} + k_{DX} \frac{S_{xn}}{|S_{xn}| + \delta} \\ U_{Yeq} &= [\lambda \ddot{e}_{yn} + \ddot{y}_D + \xi_Y S_{yn} + \eta_Y \text{sign}(S_{yn}) - w_y] \frac{m}{\mu_{2n}} \\ &\quad + k_{DY} \frac{S_{yn}}{|S_{yn}| + \delta} \quad (32) \end{aligned}$$

V. SIMULATION RESULTS

In this section the simulation results and the data used for simulation are given. The simulations are carried out in MATLAB and from the output the effectiveness of controller

is verified. The system parameter values are listed in table 1 and simulation parameter values are listed on table 2.

TABLE I. VALUES OF SYSTEM PARAMETERS

System Parameters	Values
Mass of each Drone	1.8 kg
Mass of Payload	1.8 kg
Arm length of Drone(l)	0.42m
Inertia about X axis (I_x)	2.16×10^{-3} kg. m ²
Inertia about Y axis (I_y)	2.16×10^{-3} kg. m ²
Inertia about Z axis (I_z)	0.33×10^{-3} kg. m ²

TABLE II. VALUES OF SIMULATION PARAMETERS

Simulation Parameters	Value for x,y,z tracking	Value for Φ, θ, Ψ tracking
b	5	2
ξ	5	20
η	0.1	0.1
k_D	100	20
p/q	5/7	5/7
λ	2	10
$\frac{\Delta_{Mxy}}{\Delta_F}$	1/400	1/400
δ	0.8	0.8
η	0.1	0.1
k_D	100	20
p/q	5/7	5/7
λ	2	10
Λ_{Mxy}/Λ_F	1/400	1/400
δ	0.8	0.8

Remarks 1: Since while modelling of cooperative system, it was assumed that all the quad rotors have common role in lifting the payload thus mass of payload is divided in four and added to each drone weight while carrying out the simulation. We have carried out point to point tracking in our simulation. The desired values and the initial values are considered as follows, Initial Values

Quad rotor	x	y	z	Φ	θ	Ψ
Quad rotor 1	0.1	0.0	0.0	0.35	0.4	0.65
Quad rotor 2	2.5	0.1	0.0	0.25	0.45	0.6
Quad rotor 3	2.6	2.5	0.0	0.3	0.5	0.55
Quad rotor 4	0.0	2.6	0.0	0.37	0.55	0.5

TABLE III. DESIRED VALUES(FOR ALL DRONES)

x	y	z	Φ	θ	Ψ
2	2	2	0	0	0

Figure 3, Figure 5, Figure 7 shows x, y and z displacements respectively and Figure 4, Figure 6, Figure 8 shows the errors in x, y and z direction respectively. Figure 9, Figure 11, Figure 13 shows Φ , θ and Ψ rotations respectively and Figure 10, Figure 12, Figure 14 shows the errors in roll, pitch and yaw angles respectively. From the graphs it is clearly observable that all the drone converge faster to final value with TSMC than SMC. Thus TSMC has lower settling time compared to SMC, thus it is proved to be a fast responsive system.

VI. CONCLUSION AND FUTURE WORKS

We discussed the problem of cooperative load transportation using multiple drones in three dimensions. A decentralized control law based on TSMC has been proposed. From the simulation results it is observed that system of drones has converged to desired values in finite time and thus a fast responsive system. The responses are faster in TSMC compared to conventional SMC. Also the system successfully maintained stability in the presence of wind disturbances. So the robustness of controller is verified well. The cooperative system has successfully exhibited point to point tracking. The chattering of system has been reduced to almost zero by using proper tuning parameters and discrete switching law. In future works we will be proceeding with variable point tracking and more enhanced load sharing with heavy loads.

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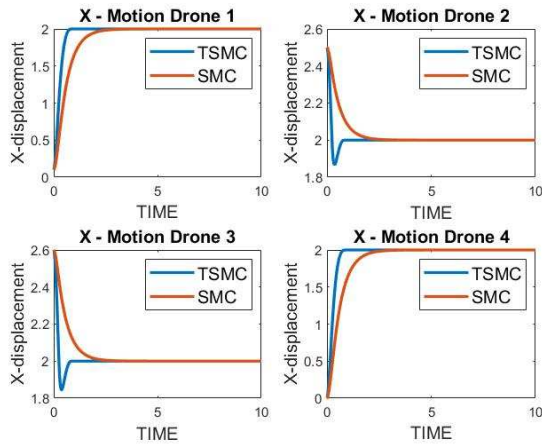


Figure 3. X motion tracking of drones

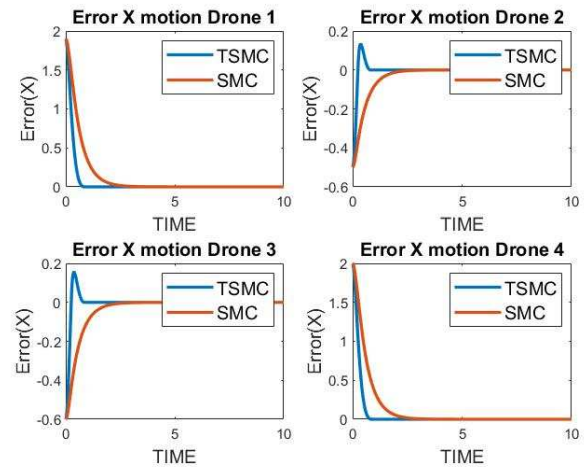


Figure 4. Error in X displacement

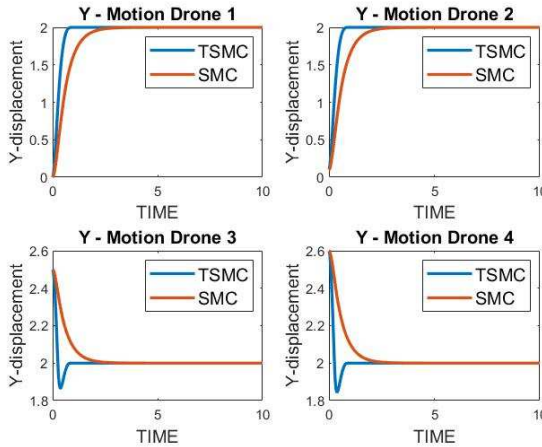


Figure 5. Y motion tracking of drones

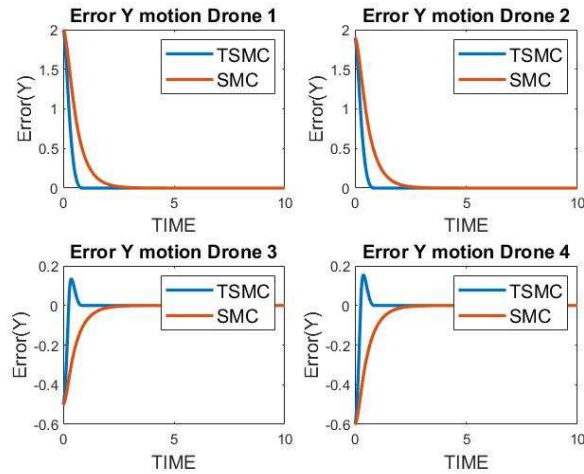


Figure 6. Error in Y displacement

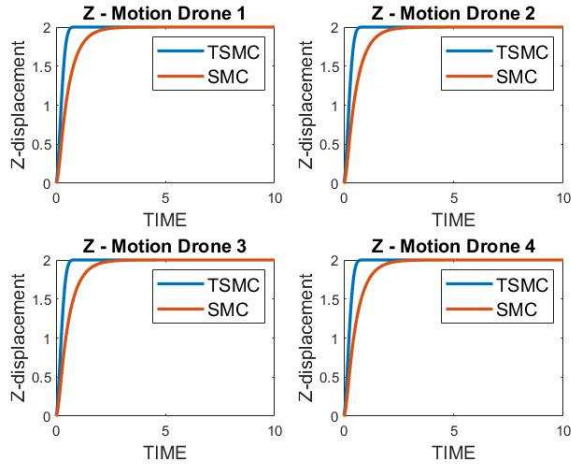


Figure 7. Z motion tracking of drones

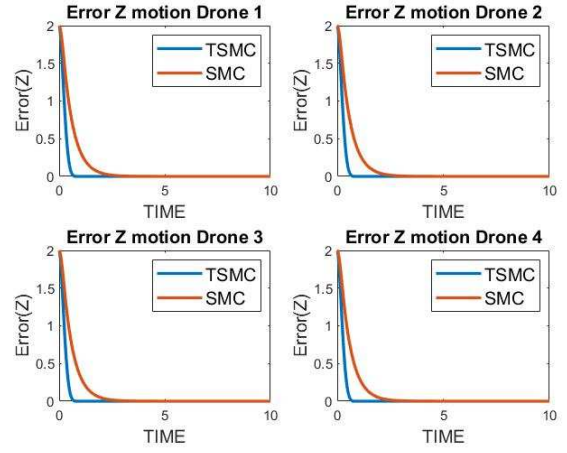


Figure 8. Error in Z displacement

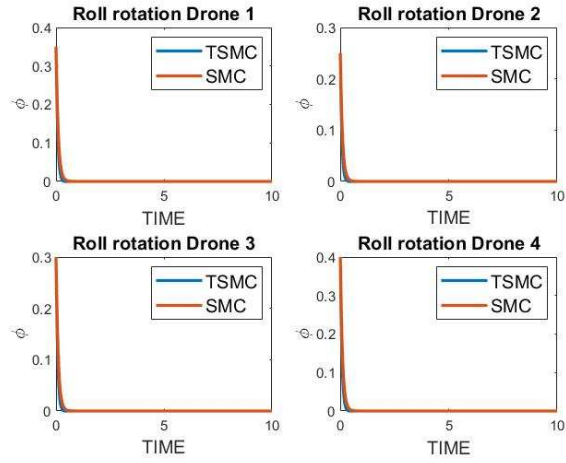


Figure 9. Roll rotation tracking of drones

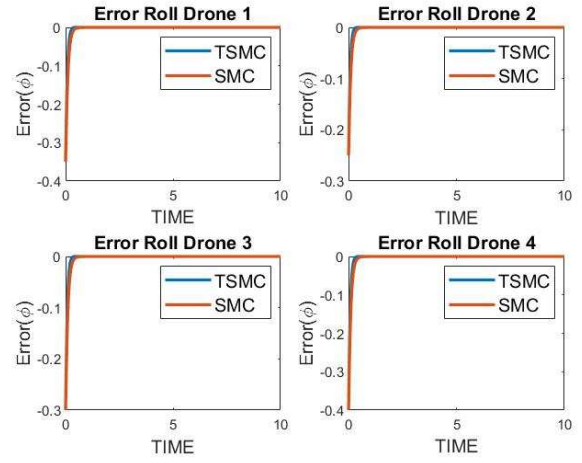


Figure 10. Error in Rolling

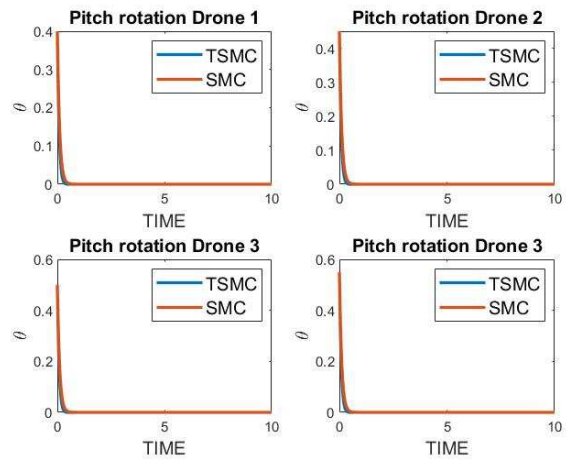


Figure 11. Pitch rotation tracking of drones

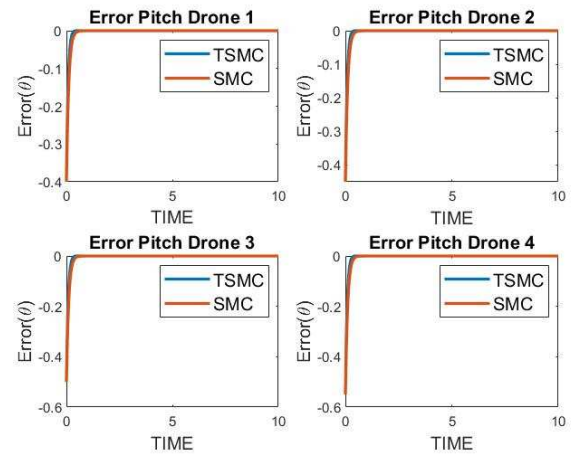


Figure 12. Error in Pitching

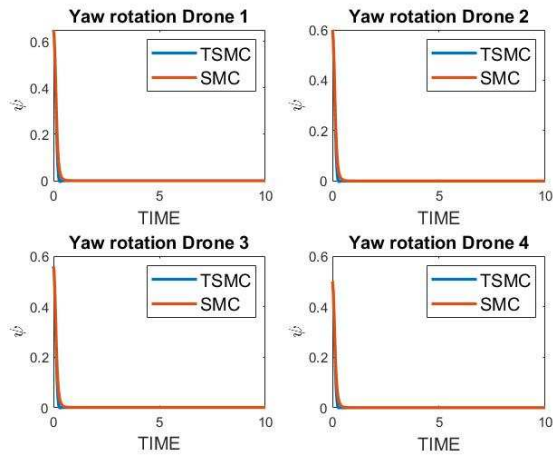


Figure 13. Yaw rotation tracking of drones

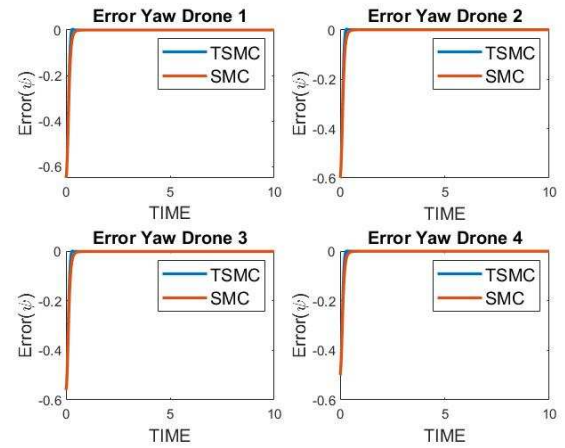


Figure 14. Error in Yawing

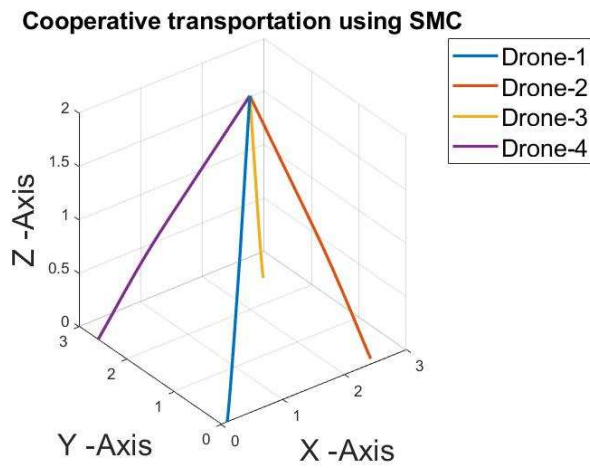


Figure 15. 3D motion tracking of drones using SMC

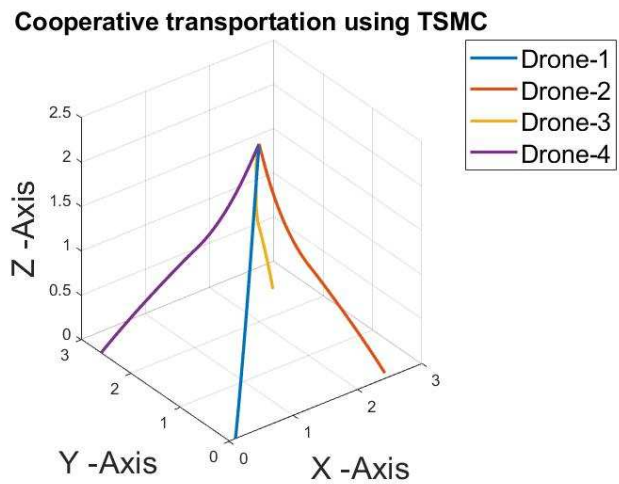


Figure 16. 3D motion tracking of drones using TSMC